An Introduction to Probability

29 October 2025

Study Guide

This week, we will be learning about:

- sample spaces, events, and the axioms of probability;
- conditional probabilities and independent events;
- random variables, expectation, and variance.

You may find it helpful to list down many examples *and* counterexamples of the various objects we encounter in this week's class. Moreover, we talk about the following key ideas:

Axiom 1. *Let S denote the sample space of an experiment. Let P be a function that associates a number $P(E)$ to each event $E \subseteq S$. Suppose that P satisfies the following axioms for each event E of S:*

1. $0 \leq P(E) \leq 1$.
2. $P(S) = 1$.
3. *If $E_1, E_2, E_3, \cdots \subseteq S$ are mutually exclusive events, then*

$$P\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} P(E_i),$$

The number $P(E)$ is the ***probability*** *of the event E.*

Theorem 2 (Bayes)**.** *Let A and B be two events. Then,*

$$P(B \mid A) = \frac{P(A \mid B)\, P(B)}{P(A)}.$$

Proposition 3 (Expectation and Variance)**.** *Let X be a discrete random variable that takes on x_i, $i \geq 1$, with respective probabilities $p(x_i)$.*

1. *For any real-valued function g,*

$$\mathbb{E}[g(x)] = \sum_i g(x_i)\, p(x_i).$$

2. *If $a, b \in \mathbb{R}$, then*

$$\mathbb{E}[aX + b] = a\mathbb{E}[x] + b, \quad \mathrm{Var}(aX + B) = a^2 \mathrm{Var}(X).$$

3. *The variance of X is given by*

$$\mathrm{Var}(X) = \mathbb{E}\left[X^2\right] - (\mathbb{E}[X])^2.$$

Definitions

Definition 4. An **experiment** is any action or process that generates observations. The set of all possible outcomes of an experiment is called its **sample space**.

Definition 5. An **event** E is any subset of an experiment's sample space. The **complement** of E, denoted E^c, is the event that E does not occur. A pair of events E and F are said to be **mutually exclusive** if they are disjoint: that is, $E \cap F = \varnothing$.

Definition 6. The **conditional probability** of E given that F has occured is defined by

$$P(E \mid F) = \frac{P(E \cap F)}{P(F)}.$$

Definition 7. A **random variable** is a real-valued function whose domain is the sample space. A random variable is said to be **discrete** if it can take on at most a countable number number of possible values. For a discrete random variable X, we define the **probability mass function** of X by

$$p(x) = P(X = x).$$

The **cumulative distribution function** of X is defined as

$$F(x) = P(X \leq x).$$

Definition 8. Let X be a discrete random variable on the sample space S with probability mass function $p(x)$. The **expected value** (or **mean**) of X, denoted by $\mathbb{E}[X]$, is given by

$$\mathbb{E}[X] = \sum_{x \in S} x \cdot p(x).$$

The **variance** of X, denoted by $\operatorname{Var}(X)$, is defined by

$$\operatorname{Var}(X) = \mathbb{E}\left[(X - \mathbb{E}[X])^2\right].$$

The square root of the variance is called the **standard deviation** of X,

$$\operatorname{SD}(X) = \sqrt{\operatorname{Var}(X)}.$$